⇒ ex3:

Express $\frac{dw}{dr}$ and $\frac{dw}{ds}$ in terms of r and s if

$w = x + 2y + z^2$, $x = \frac{r}{s}$, $y = r^2 + \ln s$, $z = 2r$

$$\frac{dw}{dr} = \frac{dw}{dx}\frac{dx}{dr} + \frac{dw}{dy}\frac{dy}{dr} + \frac{dw}{dz}\frac{dz}{dr}$$

$$= (1)\left(\frac{1}{s}\right) + (2)(2r) + (2z)(2)$$

$$\cancel{= \frac{1}{s} + 4r + 4z}$$

$$\frac{1}{s} + 4r + (4r)(2) = \frac{1}{s} + 12r$$

$$\frac{dw}{ds} = \frac{dw}{dx}\frac{dx}{ds} + \frac{dw}{dy}\frac{dy}{ds} + \frac{dw}{dz}\frac{dz}{ds}$$

$$= (1)\left(\frac{-r}{s^2}\right) + (2)\left(\frac{1}{s}\right) + (2z)\,0 = \frac{2}{s} - \frac{r}{s^2}$$

→ Note: Multivariable chain Rule (3 variables, 2 perameters.)

if:

$w = w(x, y, z)$, $x = x(r, s)$, $y = y(r, s)$, $z = z(r, s)$

then

~~for~~ $\cancel{dw}/dr$:

for $\frac{dw}{dr}$:

$$\frac{dw}{dr} = \frac{dw}{dx}\frac{dx}{dr} + \frac{dw}{dy}\frac{dy}{dr} + \frac{dw}{dz}\frac{dz}{dr}.$$

→ Note :- Prevous formulas

- w depends on two variables x and y.
- x, y depend on one variable t.

~~But~~ $$\boxed{\frac{dw}{dt} = \frac{dw}{dx}\frac{dx}{dt} + \frac{dw}{dy}\frac{dy}{dt}}$$

But in Eqn:-3

- w depends on three variables, x, y, z.
- x, y, z depend on two variables r and s.